Digital Data Transmission 2026–27

Universitat Pompeu Fabra Barcelona

Modality A: Midterm exam (sample 2)

- Each statement below concerns one part of a statement of the Midterm collection, and it may be worded differently from the collection. Clearly **select** (a) if the statement is true and (b) if it is false. Marks are +1 for a correct answer, −1 for a wrong answer, and 0 for no answer.
- No derivations or justifications are required, and none is marked.
- The exam **grade** is $R - W$, the number of right answers minus the number of wrong answers, over the 12 statements.
- No electronic devices and no printed or written material are allowed.
- The exam duration is **30 minutes**.

Name:

NIA or U:

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1. A set $\{\phi_i(t)\}$ of waveforms is orthonormal if $\langle \phi_i, \phi_k \rangle = \delta_{ik}$: every waveform has unit energy and any two distinct waveforms are orthogonal.

(a) True
(b) False

2. By Parseval's identity, an inner product can be computed in the time or in the frequency domain,

$$\langle x, y \rangle = \int_{-\infty}^{\infty} \hat{x}(f)\hat{y}^*(f)\,\mathrm{d}f,$$

so a signal has the same energy in both domains.

(a) True
(b) False

3. A rectangle of amplitude A and width T centered at the origin has Fourier transform $A\,\mathrm{sinc}(Tf)$.

(a) True
(b) False

4. A baseband waveform of bandwidth 50 MHz allows at most 100 Msymbol/s and 800 Mbit/s with 256-QAM.

(a) True
(b) False

5. For a complex signal, $|x(t)|^2 = x(t)x^*(t)$ is the squared modulus, not the square, and the energy is $\int_{-\infty}^{\infty} |x(t)|^2\,\mathrm{d}t$.

(a) True
(b) False

6. The average energy per bit is $E_\mathrm{b} = E_\mathrm{s} \log_2 M$ for a one-to-one mapping of $\log_2 M$ bits onto M symbols.

(a) True
(b) False

7. In a Monte Carlo simulation, the relative frequency of an event approaches its probability, because the indicator of the event, equal to 1 when it occurs and 0 otherwise, has expectation equal to the probability.

(a) True
(b) False

8. The entropy of a discrete random variable is nonnegative, and it is additive for independent variables.

(a) True
(b) False

9. A real Gaussian random variable with variance σ^2 has differential entropy $\frac{1}{2}\log_2(2\pi e\sigma^2)$, which depends on its mean.

(a) True
(b) False

10. The mutual information $I(X;Y) = H(Y) - H(Y|X)$ is nonnegative but not symmetric: in general $I(X;Y)$ differs from $I(Y;X)$.

(a) True
(b) False

11. As the bandwidth grows without bound with P and N_0 fixed, the capacity of the AWGN channel grows without bound as well.

(a) True
(b) False

12. On the Gaussian channel the ML detector chooses the constellation point nearest to the observation, and this nearest-point rule also holds with uniform, exponential or multiplicative noise.

(a) True
(b) False